A Toy Demonstration of Quantum Singular Value Transformation on the 1D Poisson Equation

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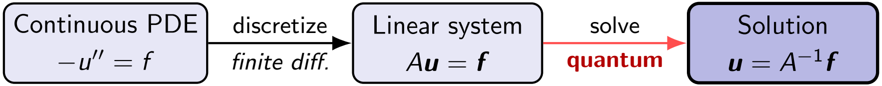
Demonstrating the QSVT mechanism on a tiny finite-difference system

The General Recipe

Solving a PDE numerically almost always follows the same path:



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Solving a PDE numerically almost always follows the same path:

$$\boxed{\begin{array}{c}\text{Continuous PDE}\\ -u'' = f\end{array}} \xrightarrow[\textit{finite diff.}]{\text{discretize}} \boxed{\begin{array}{c}\text{Linear system}\\ A\boldsymbol{u} = \boldsymbol{f}\end{array}} \xrightarrow[\textbf{quantum}]{\text{solve}} \boxed{\begin{array}{c}\text{Solution}\\ \boldsymbol{u} = A^{-1}\boldsymbol{f}\end{array}}$$

- **Steps 1–2 (classical):** mesh the domain and replace derivatives by finite differences $\Rightarrow$ a matrix equation $A\boldsymbol{u} = \boldsymbol{f}$.
- **Step 3 (quantum):** the matrix equation is where QSVT enters — block-encode A and apply a polynomial inverse to get $A^{-1}\boldsymbol{f}$.

This talk: a two-point Poisson example carried through every step.

The Problem: 1D Poisson Equation

We solve the boundary-value problem

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With two interior points the tridiagonal Laplacian collapses to a 2×2 system:

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \qquad \boldsymbol{f} = \begin{pmatrix} 1 \\ 0.3 \end{pmatrix}.$$

Goal: produce the normalized state $|u\rangle \propto A^{-1}\,|f\rangle$ with a quantum circuit.

The QSVT Idea — Why $W(A)$, Why a Polynomial

A quantum circuit is *unitary*, but A is not. Two ideas bridge the gap.
1. Block-encoding. Rescale A to $\|A\| \leq 1$ and hide it as the top-left block of a unitary — the *signal unitary*:

$$W(A) = \begin{pmatrix} A & i\sqrt{I - A^2} \\ i\sqrt{I - A^2} & A \end{pmatrix}, \qquad (\langle 0| \otimes I)\, W(A)\, (|0\rangle \otimes I) = A.$$

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2. QSVT. Interleaving W with ancilla phase rotations $S(\phi) = e^{i\phi Z}$ applies a *polynomial to the eigenvalues* of A:

$$U_\Phi = S(\phi_0)\, W\, S(\phi_1)\, W \cdots W\, S(\phi_d), \quad [U_\Phi]_{\text{top-left}} = P(A) = \sum_j P(\lambda_j)\, |\lambda_j\rangle\langle\lambda_j|\,.$$

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Inversion. QSVT sends each eigenvalue $\lambda_j \to P(\lambda_j)$. Since A^{-1} inverts every eigenvalue, choosing $P(x) \approx \gamma/x$ on the spectrum builds $P(A) \approx \gamma\, A^{-1}$. So $P(A)\,|f\rangle \propto A^{-1}\,|f\rangle$ — the solution — kept by postselecting the ancilla on $|0\rangle$.

Implementation for the Two-Point Poisson

On the eigenvalues $\{\frac{1}{4}, \frac{3}{4}\}$ of $B = A/4$, approximate γ/x ($\gamma = \frac{3}{32}$) by the exact degree-3 polynomial

$$p(x) = \frac{5}{3}\,x - \frac{8}{3}\,x^3.$$

The four phases realizing p, and their sequence:

$$\Phi = [0.0082, -0.6155, 0.6155, 3.1334], \qquad U_\Phi = S(\phi_0)W(B)S(\phi_1)W(B)S(\phi_2)W(B)S(\phi_3).$$

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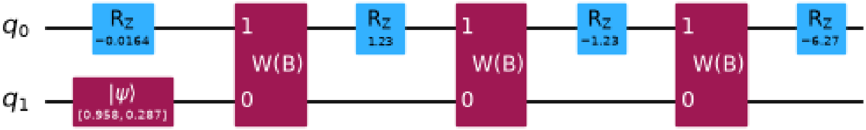
Realized as a circuit ($S(\phi) \to R_z(-2\phi)$, system q_1 prepared as $|f\rangle$):



Result

Classical reference. Solve $A\boldsymbol{u} = \boldsymbol{f}$ with ordinary numerical linear algebra (`numpy`), then normalize — the quantum output carries no overall scale:

$$\boldsymbol{u}/\|\boldsymbol{u}\| = (0.8209,\ 0.5711).$$

Quantum output. Postselect the ancilla on $|0\rangle$ and normalize the two circuit amplitudes:

$$(0.8209,\ 0.5711).$$

The two normalized vectors agree componentwise to $\sim 10^{-9}$, set by the rounded phase angles. Postselection succeeds with probability ≈ 0.11.

Summary & Outlook

Done: block-encode A, apply an exact inverse polynomial via a degree-3 QSVT phase sequence, run as a Qiskit circuit — its output matches the classical solution to $\sim 10^{-9}$.

Hard at scale: state preparation of $|f\rangle$, phase-angle synthesis for general polynomials, and higher degrees as the spectrum widens.

Refs: Lin, arXiv:2201.08309 (§8.4); Childs–Liu–Ostrander, arXiv:2002.07868; Shaviner–Chen–Frankel, arXiv:2507.09686.

Thank you!